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CSC2001F Assignment 1

Report:

### OO Design:

Classes created:

- Facts.java - Class to store the key phrase, factual statement, and confidence level of each fact in the knowledge base. This is the main object.
- FactsArray.java - Class to manage an array of Facts objects and provide functionality, including adding facts from a text file or adding facts individually to the array, updating facts, searching for facts by key and by key and factual statement.
- GenericsKbArrayApp.java - Provides an interface for the user to interact with the knowledge base array. This class allows the user to utilize the functionality provided by the Facts Array class, as they select which task they would like to perform (of the 4 options) and then the class uses the Facts Array to perform these tasks.
- BinaryTreeNode.java – provides the node functionality for the binary tree class, by referencing the order of nodes (Facts objects) in the tree.
- FactsBST.java – provides the same functions as the Facts Array class but in the form of a binary search tree rather than an array. The functions include adding facts from a text file or adding facts individually to the BST, updating facts, searching for facts by key and by key and factual statement.
- GenericsKbBSTApp.java - Provides an interface for the user to interact with the knowledge base BST. This class allows the user to utilize the functionality provided by the Facts BST class, as they select which task they would like to perform (of the 4 options) and then the class uses the Facts BST to perform these tasks.

I chose this arrangement of classes as it simplifies the creation and understating of the code at each stage, the use of the Facts object class allows for a single creation of the class to be used in multiple more advanced classes. The separation of the node class and BST classes allow for code re-use as well. And finally the 2 UI classes allow for a clear distinction between which data structure type is being used based on the UI class selected.

Testing process:

Text file used:

Test1.txt:

```
Test1.txt
File Edit View

carious dentin Carious dentin is made up of two distinct layers. 0.7570869326591492
travel agency Travel agencies are agencies. 1.0
lobule Lobules consist of groups of acini forming grape-like structures. 0.8180095553398132
third grade science Third grade science relate to scientific investigations. 1.0
ocelot Ocelots thrive in a variety of habitats, from lush groves to arid deserts and forests. 0.7324625849723816
sea water Sea water has capacity. 1.0
diazepam Diazepam is benzodiazepines. 1.0
ethylene glycol Ethylene glycol is produced commercially almost exclusively by the hydration of ethylene oxide. 0.8011720776557922
seizure Seizures are a symptom of epilepsy. 0.7180448770523071
writing Writing brainstorm. 1.0
prednisone Prednisone irritates the adrenal glands and other organs. 0.814678430557251
chloral hydrate Chloral hydrates are chemical compounds. 1.0
bacterial cell Bacterial cells can contain thousands of ribosomes , the sites of protein synthesis. 0.8179494142532349
humanity Humanity has (part) cell membranes. 1.0
mount A mount is a horse 1.0
salt Salt is used for seasons. 1.0
air bubble Air bubbles are located in water. 1.0
metrology A metrology is science 1.0
folk art Folk art is genres. 1.0
leatherback sea turtle Leatherback sea turtles have carapaces. 1.0
feed tadpole Feed tadpoles become eat frogs. 1.0
gradient wind Gradient wind consists of forces. 1.0
sound barrier A sound barrier is drag 1.0
quick decision Quick decisions base on information. 1.0
rightist A rightist is a conservative 1.0
other important issue Other important issues relate to treatments. 1.0
tear gas Tear gases are gases. 1.0
eider duck Eider ducks use their down to line their nests and cover their eggs. 0.7294564247131348
structural adhesive Structural adhesive has strong chemical resistance. 1.0
caliche A caliche is a stone 1.0
modern reproductive technology Modern reproductive technology has effects. 1.0
cooper A cooper is an artisan 1.0
lunching Lunchings is eating. 1.0
fern Ferns have large leaves with branching vascular tissue. 0.8390014171600342
mutilation A mutilation is an injury 1.0
hepatitis c treatment Hepatitis c treatments cause side effects. 1.0
gross sale Gross sales are incomes. 1.0
kimono A kimono is a robe 1.0
lister A lister is an assessor 1.0
heavy water Heavy water is water with a deuterium atom instead of one of the hydrogen atoms. 0.8067517280578613
tapir Tapirs have (part) ears. 1.0
electric battery An electric battery is an electrical device 1.0
```

Generics UI testing:

Array testing:

Choose an action from the menu:

1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit

Enter your choice:

▶▶ 1

```
Enter file name:
▶▶ Test1.txt
Loaded successfully

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 1
Enter file name:
▶▶ Fake.txt
unable to load knowledge base

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 2
Enter the term:
▶▶ Mist
Enter the statement:
▶▶ Mist is water vaper in the air
Enter the confidence score:
▶▶ 0.75
Statement added successfully

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 2
Enter the term:
```

```
▶▶ Mist
Enter the statement:
▶▶ Mist often forms when warmer air over water suddenly encounters
the cooler surface of land.
Enter the confidence score:
▶▶ 1
Statement updated successfully

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 2
Enter the term:
▶▶ Mist
Enter the statement:
▶▶ Mist is more dense than water
Enter the confidence score:
▶▶ 0.3
Statement could not be added

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 3
Enter the term to search:
▶▶ Sea water
Statement found: Sea water Sea water has capacity.. (Confidence
score: 1.0)

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
```

4. Search for a item in the knowledge base by term and sentence
5. Quit

Enter your choice:

▶▶

3

Enter the term to search:

▶▶

Density

Term not found

Choose an action from the menu:

1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit

Enter your choice:

▶▶

4

Enter the term to search:

▶▶

Mist

Enter the statement to search for:

▶▶

Mist often forms when warmer air over water suddenly encounters the cooler surface of land.

The statement was found and has a confidence score of 1.0.

Choose an action from the menu:

1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit

Enter your choice:

▶▶

4

Enter the term to search:

▶▶

Sky

Enter the statement to search for:

▶▶

The sky is purple on hot days

Term not found

Choose an action from the menu:

1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term

```

4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 4
Enter the term to search:
▶▶ Sea water
Enter the statement to search for:
▶▶ sea water is healthy to drink
Term not found

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 7
Invalid input

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 5
Goodbye!

```

## BST testing:

```

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 1
Enter file name:

```

```
▶▶ Test1.txt
Loaded successfully

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 1
Enter file name:
▶▶ NotAFile.txt
unable to load knowledge base

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 2
Enter the term:
▶▶ RDLRUT001
Enter the statement:
▶▶ This is my student number
Enter the confidence score:
▶▶ 1
Statement added successfully

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 2
Enter the term:
▶▶ Assignment1
```

Enter the statement:

▶▶ due sate is the 6th of March

Enter the confidence score:

▶▶ 0.5

Statement added successfully

Choose an action from the menu:

1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit

Enter your choice:

▶▶ 2

Enter the term:

▶▶ Assignment1

Enter the statement:

▶▶ due date is the 8th of March

Enter the confidence score:

▶▶ 0.9

Statement updated successfully

Choose an action from the menu:

1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit

Enter your choice:

▶▶ 2

Enter the term:

▶▶ Assignment1

Enter the statement:

▶▶ I will not complete the assignment on time

Enter the confidence score:

▶▶ 0.005

Statement could not be updated or added

Choose an action from the menu:

1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term



```

4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
>> 3
Enter the term to search:
>> rightist
Statement found: rightist A rightist is a conservative.
(Confidence score: 1.0)

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
>> 3
Enter the term to search:
>> Clouds
Statement not found

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
>> 4
Enter the term to search:
>> RDLRUT001
Enter the statement to search for:
>> This is my student number
Statement found: RDLRUT001 This is my student number. (Confidence
score: 1.0)

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence

```

```
5. Quit
Enter your choice:
▶▶ 4
Enter the term to search:
▶▶ sea water
Enter the statement to search for:
▶▶ sea water is safe to drink
Statement not found

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 25
Invalid input

Choose an action from the menu:
1. Load a knowledge base from a file
2. Add a new statement to the knowledge base
3. Search for an item in the knowledge base by term
4. Search for a item in the knowledge base by term and sentence
5. Quit
Enter your choice:
▶▶ 5
Goodbye!
```

## Creativity:

I added an option for if the user enters an integer that is not a valid selection, the user is informed and asked to enter a valid choice. I also chose to keep the update functionality in both sets of UI systems as I felt this enhanced the program substantially.

## Git log:

0: commit faaf502f43311b13ab3149ac78ad95372f3ca762

1: Author: Ruth Rudolph <RDLRUT001@myuct.ac.za>

2: Date: Sun Mar 3 18:19:58 2024 +0200

3:

4: Final edits and Javadoc's completed, ready to submit the code at this stage

5:

6: commit db0289951ff9e0c1ead37cd9bc7234c331dbd67e

7: Author: Ruth Rudolph <RDLRUT001@myuct.ac.za>

8: Date: Sun Mar 3 11:48:50 2024 +0200

9:

...

49: Author: Ruth Rudolph <RDLRUT001@myuct.ac.za>

50: Date: Wed Feb 28 17:37:11 2024 +0200

51:

52: Created a frame for the manager class which will perform the functions involving the array and return values for the output to the UI class

53:

54: commit 0de88f11c1c495934e2ed0a1c2a78c2402d3a2ee

55: Author: Ruth Rudolph <RDLRUT001@myuct.ac.za>

56: Date: Wed Feb 28 17:22:05 2024 +0200

57:

58: Created a 'Facts' object class to store the details of an item from the knowledge base